

Crime Reporting and Public Safety Management System

Introduction

With the increasing number of crime incidents, traditional crime reporting systems often face several challenges such as delays, lack of transparency, and limited accessibility. Many citizens hesitate to report crimes due to complicated procedures, lack of awareness, or fear of visiting police stations.

To address these issues, this project presents a web-based Crime Reporting and Public Safety Management System that allows users to easily file complaints online by submitting details like location, date, description, and evidence.

The platform includes a centralized dashboard for authorities to view, monitor, and update case statuses in real time. It also features automated complaint classification based on user descriptions to help organize and prioritize cases effectively.

Overall, the system aims to improve communication between citizens and law enforcement agencies, enhance transparency in complaint handling, and contribute to better public safety management.

Objectives

Enhance Crime Reporting Accessibility

- The primary objective of this project is to provide an easy-to-use online platform that allows citizens to report crimes conveniently from anywhere. This reduces dependency on physical visits to police stations and encourages more people to report incidents.

Automate Crime Classification

- The project focuses on automatically identifying the type of crime based on the user's description using machine learning techniques. This helps in organizing complaints and assists authorities in quicker decision-making.

Improve Transparency and Case Tracking

- Another objective is to provide users with the ability to track the status of their complaints using a unique tracking ID. This increases transparency and builds trust between citizens and law enforcement.

Support Police Decision-Making

- The system provides a dashboard for police officers to view, filter, and update complaints based on various factors such as location, date, and status. This improves monitoring and response efficiency.



Methods and Techniques

Methods

1. Data Collection

- A publicly available crime dataset is used containing crime categories and descriptions.
- The dataset includes textual descriptions of incidents and their corresponding crime types.
- This data is used to train the classification model.

2. Data Preprocessing Techniques

- Removal of missing or null values
- Conversion of text to lowercase
- Removal of special characters and numbers
- Tokenization of text data
- Removal of stopwords

These steps help in cleaning the data and improving model performance.

3. Feature Extraction Technique

TF-IDF (Term Frequency–Inverse Document Frequency) is used.

Converts textual data into numerical vectors

Highlights important words in crime descriptions

4. Machine Learning Technique

Naive Bayes Classifier is used for classification

Suitable for text classification problems

Fast, efficient, and performs well on large text datasets

5. System Development Approach

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js with Express
- Database:MySQL
- ML Integration: Python Flask API

The system connects user input → backend → ML model → database → dashboard.

6. Training and Evaluation Techniques

80–20 Train-Test Split

Model trained using labeled crime data

Performance Metrics: Accuracy, Precision, Recall, F1-score

Algorithm

Step1:LoadDataset

Import crime dataset containing descriptions and crime types.

Step 2: Preprocess Data

Clean text by removing noise, converting to lowercase, and filtering unnecessary characters.

Step 3: Feature Extraction

Convert cleaned text into numerical vectors using TF-IDF.

Step 4: Train Model

Train Naive Bayes classifier using processed data.

Step 5: Evaluate Model

Test model performance using evaluation metrics such as accuracy, precision, recall, and F1-score.

Step 6: Deploy Model

Save trained model and integrate it with a Flask API.

Step 7: User Complaint Submission

User submits complaint through web interface.

Step 8: Crime Classification

Backend sends complaint description to ML model → predicted crime type is returned

Step 9: Store Data

Complaint along with predicted crime type is stored in MySQL database.

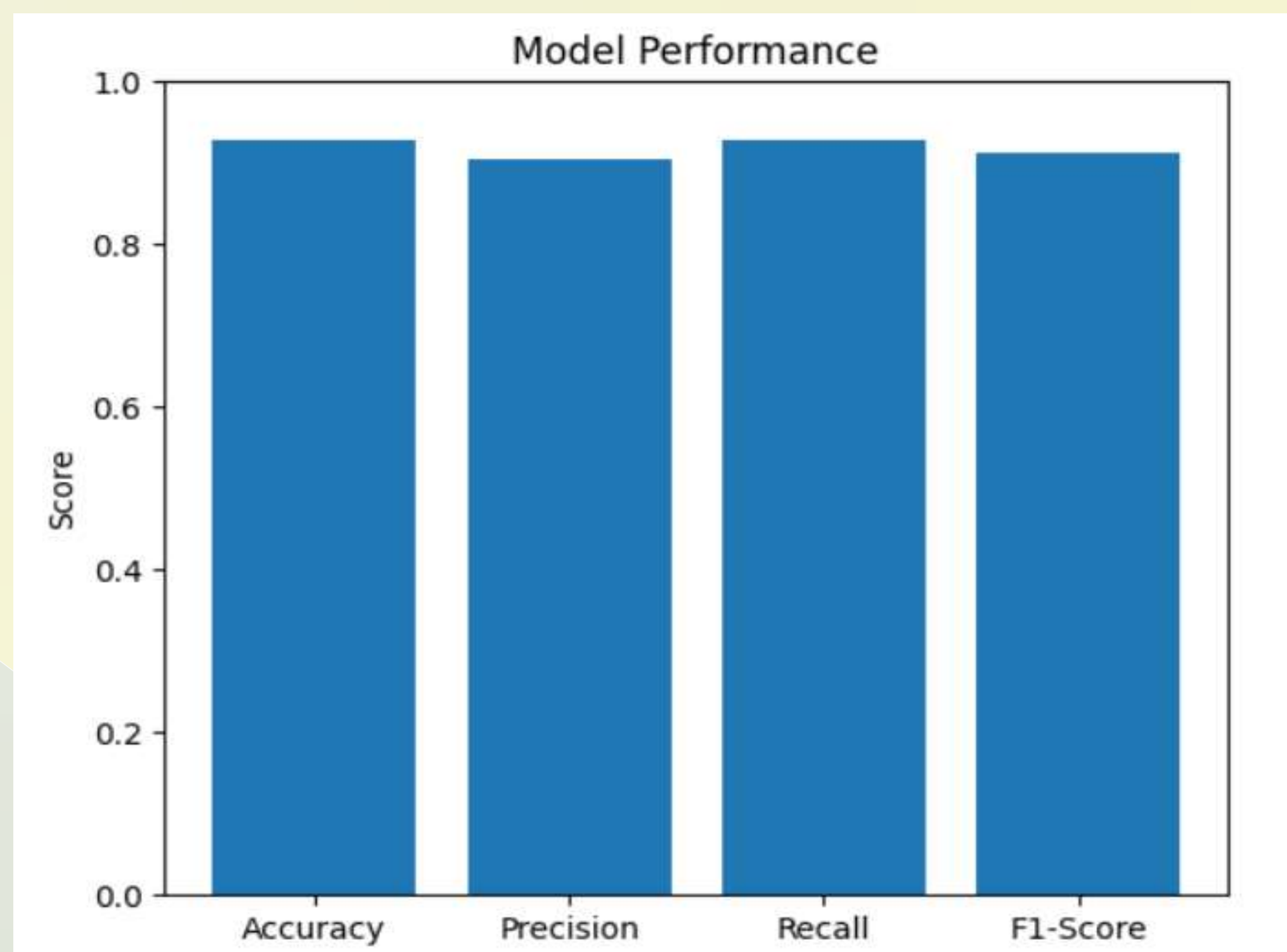
Step 10: Display Results

User can track complaint status

Police dashboard displays all complaints with filtering and updates.

Results

- The system successfully enables users to report crimes online in a simple and accessible way
- Complaint data is securely stored in the database and can be retrieved in real time.
- Submitted complaints are automatically processed and classified into crime categories.
- Each complaint is assigned a unique tracking ID for monitoring case progress.
- Status updates made by police are reflected instantly in both dashboard and user tracking page.
- The overall system improves transparency, reduces manual effort, and speeds up complaint handling.



Police Dashboard				
dd-mm-yyyy All Crime Types v All Status v Apply Filter Apply Filter Reset Reset				
ID	Place	Crime Type	Status	Action
5	hyderabad	LARCENY/THEFT	Pending	Update v
6	narsapur	NON-CRIMINAL	Solved	Update v
7	secunderabad	LARCENY/THEFT	Solved	Update v
8	secunderabad	VEHICLE THEFT	Pending	Update v
9	warangal	ASSAULT	Solved	Update v
10	warangal	NON-CRIMINAL	Pending	Update v

Your Complaints				
ID	Name	Place	Crime Type	Status
7	radha	secunderabad	LARCENY/THEFT	Solved
8	radha	secunderabad	VEHICLE THEFT	Pending
9	radha	warangal	ASSAULT	Solved
10	radha	warangal	NON-CRIMINAL	Pending

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Conclusions

The Crime Reporting and Public Safety Management System provides an efficient and user-friendly solution for modern crime reporting. The system allows citizens to submit complaints online, reducing the need for physical visits to police stations and making the process more accessible.

By integrating automated crime classification, secure data storage, and a centralized dashboard for police, the system improves the speed and accuracy of handling complaints. Users can track their cases in real time, while law enforcement can manage and update cases effectively.

Overall, the system enhances transparency, reduces manual effort, and improves coordination between citizens and police authorities. It serves as a scalable and practical solution for improving public safety and digital governance.

References

- [1] R. Kumar et al., “Online Crime Reporting System,” 2024.
- [2] Patel et al., “E-Crime Reporting and Management System,” 2024.
- [3] Sharma et al., “Crime Data Management System,” 2025.

SDGs: 16,9

SDG 16 – Peace, Justice, and Strong Institutions

Promotes transparent, efficient crime reporting and strengthens trust between citizens and law enforcement.

SDG 9 – Industry, Innovation, and Infrastructure

Encourages digital innovation through a smart, scalable crime reporting platform.

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